DMMM IAE I

1. **Define & differentiate between Disasters & Hazard.**

1. Disaster

A **disaster** is a sudden, catastrophic event that seriously disrupts the functioning of a community or society, causing widespread human, material, economic, or environmental losses that exceed the community's ability to cope using its own resources.

Example: Earthquakes, floods, industrial explosions.

2. Hazard

A **hazard** is a dangerous condition or event that has the potential to cause injury, loss of life, damage to property or the environment.

Types:

- **Natural hazards** – Earthquakes, cyclones, floods
- **Man-made hazards** – Industrial accidents, chemical spills, nuclear leaks

Example • Biological Hazards: cholera, dengue-fever, Malaria Tuberculosis etc.
• Meteorological: cyclone, storms, Tidal waves, Hurricane etc.
• Technological Hazards: Transport Accident.

Diffrence between Hazard and disaster

Hazard	Disaster
- A Hazard is a perceived natural-event which threatens life, property and environment	- A disaster is consequence of hazard which occurs when hazard meets vulnerable situation leads to great loss of life, prop and e-commerce
- community and e-commerce is able to cope with hazard using available resource	- A community is overwhelmed by hazard and is unable to manage with current resources
- Initial - hazard is not considered as disaster	- Hazard leads to disaster by creating vulnerable situation

2. **Define & differentiate between Vulnerability & Risk.**

Definition:

1. Vulnerability:

Vulnerability refers to the **degree to which a system, community, or individual is likely to experience harm due to** the **exposure to a hazard.**

It is influenced by factors like:

- **Poverty**
- **Poor infrastructure**
- **Lack of awareness or preparedness**
- **Social and economic conditions**

✅ **Example:** A poorly constructed house in an earthquake-prone area is **highly vulnerable** to damage.

2. Risk:

Risk is the **potential for loss or damage when a hazard interacts with vulnerable conditions.**

It is typically expressed as:

Risk = Hazard × Vulnerability × Exposure

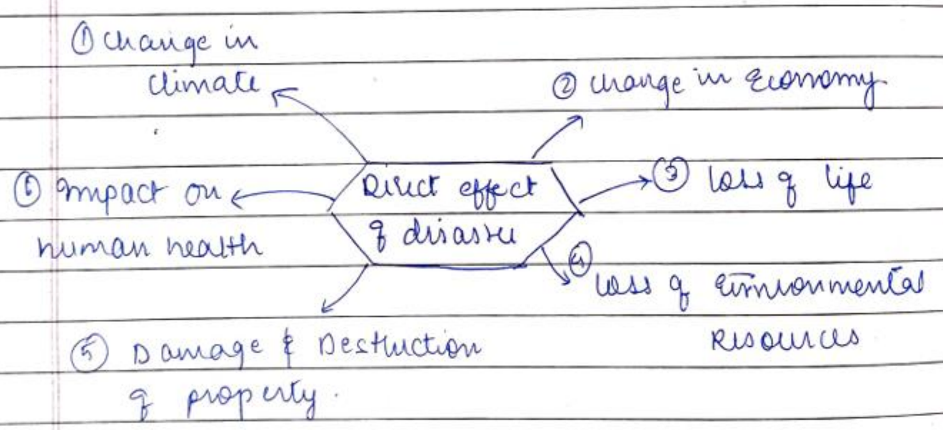
✅ **Example:** If a flood-prone area has high population density and poor drainage systems, the **risk of flood damage** is high.

Key Differences:

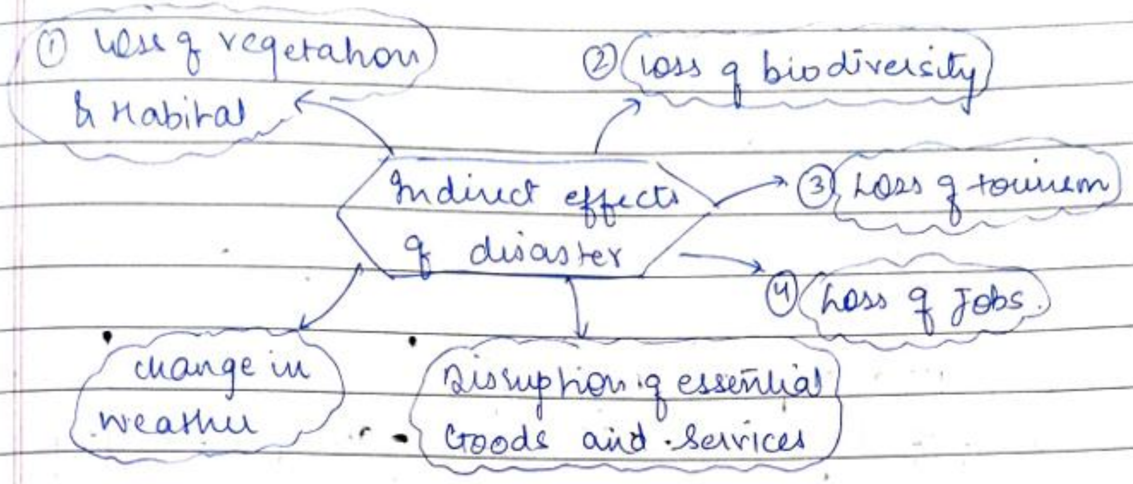
Aspect	Vulnerability	Risk
Definition	Susceptibility to damage from hazards	Potential for loss or damage due to hazard impact
Depends on	Social, economic, physical, and environmental factors	Combination of hazard, vulnerability, and exposure
Role	Component of risk	Overall measurement of potential impact
Example	A village with no early warning system (vulnerable)	Risk of casualties if a cyclone hits that village

3. Discuss the direct & indirect effects of disasters with one example for each.

- Direct effects of disaster refers to direct quantifiable losses that is physical and structural impact caused by the disaster such as damage to property and destruction of property of cultural significance.



- Indirect effects of disaster refers to indirect non-quantifiers losses that is non-physical impact caused by the disaster such as disruption of essential goods and services.



4. **What is climate change? What are the effects or impacts of Global Warming**

Climate Change

Climate change refers to long-term shifts in temperature, weather patterns, and climatic conditions on Earth, mainly caused by **natural processes** (like volcanic eruptions) and **human activities** (like burning fossil fuels, deforestation, industrialization).

It results in **global warming, rising sea levels, extreme weather events, and ecosystem disruption**.

Effects of Global Warming

1. **Loss of Biodiversity** – Rise in temperature affects ecosystems; many plants and animals cannot withstand heat and die.
2. **Sea Level Rise** – Melting of ice in Arctic and Antarctic regions causes sea levels to rise, submerging coastal areas and islands (e.g., Maldives at risk).
3. **Climatic Changes** – Changes in seasons and irregular rainfall; extreme heat waves (e.g., >40°C in Maharashtra, 2019), impacting human life physically and economically.

5. Define & explain (Write a short note on):

(i) Cloud burst

1. **Definition (IMD):** A cloud burst involves **rainfall of 10 cm (100 mm) or more within an hour** over a small area of about **10–30 sq. km**.
2. **Nature:** It is a **sudden, localized, and intense rainfall** event.
3. **Impact:** Causes **flash floods, landslides, and severe damage** to life, property, and agriculture.
4. **Occurrence:** Mostly happens in **hilly and mountainous regions** due to rapid condensation of moisture-laden clouds.

(ii) Volcanic eruption.

1. **Definition:** A volcanic eruption is the **sudden release of molten rock (lava), gases, and ash** from an opening in the Earth's crust.
2. **Cause:** It occurs due to **movement of tectonic plates** and pressure build-up of magma beneath the surface.
3. **Impact:** Leads to **loss of life, destruction of property, air pollution, climate disturbances**, and formation of new landforms.
4. **Examples:** Famous eruptions include **Mount Vesuvius (Italy), Mount Krakatoa (Indonesia), and Mount St. Helens (USA)**.

6. Define & differentiate between:

(i) Storm & Storm-Surge

Point	Storm	Storm Surge
1. Definition	A storm is a **violent atmospheric disturbance** characterized by strong winds, heavy rainfall, and sometimes thunder/lightning.	A storm surge is an **abnormal rise of sea level** caused by strong winds and low pressure during a cyclone/storm.
2. Cause	Caused by **atmospheric conditions** like pressure differences, cyclones, or depressions.	Caused by **cyclones and strong winds** pushing seawater towards the coast, combined with low atmospheric pressure.
3. Nature	Involves **wind, rain, thunder, and turbulence** in the atmosphere.	Involves **coastal flooding and seawater intrusion** onto land.
4. Impact	Leads to **heavy rainfall, flooding, crop damage, and infrastructure loss**.	Leads to **widespread coastal flooding, property destruction, salinity intrusion, and loss of life**.
5. Examples/Occurrence	Common in regions affected by **cyclones, thunderstorms, hurricanes, typhoons**.	Specifically occurs **along coastal areas** during cyclones, especially near the cyclone's center.

(ii) Cyclone & Cyclone-Surge.

Difference between Cyclone and Cyclone Surge

Point	Cyclone	Cyclone Surge
1. Definition	A cyclone is a **large-scale weather system** with low pressure at the center, characterized by strong winds spiraling inward and heavy rainfall.	A cyclone surge (storm surge caused by cyclone) is an **abnormal rise in sea level** during a cyclone due to strong winds and low atmospheric pressure.
2. Nature	It is a **meteorological phenomenon** involving wind circulation and rainfall.	It is an **oceanographic phenomenon** involving sea water being pushed towards coastal areas.
3. Cause	Caused by **low-pressure systems** and movement of air masses over warm ocean waters.	Caused by **cyclone winds pushing seawater** towards the coast, combined with low pressure.
4. Impact	Leads to **destruction by winds, heavy rainfall, floods, and infrastructure damage**.	Leads to **coastal flooding, saltwater intrusion, property damage, and loss of life**.
5. Example/Occurrence	Examples: Cyclone Fani (2019), Cyclone Amphan (2020).	Occurs along **coastal areas hit by cyclones**, especially near the landfall point.

7. Classify various types of disasters.

1. **Natural Disasters**
   - Caused by natural forces.
   - Examples: Earthquakes, cyclones, floods, droughts, volcanic eruptions, tsunamis, landslides.
2. **Man-made (Anthropogenic) Disasters**
   - Result from human activities or negligence.
   - Examples: Industrial accidents, chemical spills, nuclear accidents, deforestation, pollution, wars, terrorism, fire accidents.
3. **Technological Disasters**
   - Failures of technology or infrastructure.
   - Examples: Industrial explosions, power grid failures, transport accidents (road, rail, air, sea).
4. **Biological Disasters**
   - Caused by living organisms or biological processes.
   - Examples: Epidemics, pandemics (COVID-19), pest infestations, animal attacks.
5. **Environmental Disasters**
   - Caused by long-term environmental degradation.
   - Examples: Climate change, desertification, loss of biodiversity, ozone depletion.

8. Write about any four disaster-risks in India.

1. **Earthquakes**
   - India lies in a seismically active zone, with the **Himalayan region and North-East states** being highly prone.
   - Major quakes (e.g., Bhuj 2001, Kashmir 2005) caused severe destruction of life and property.
2. **Floods**
   - Annual monsoon rains often cause floods in **Assam, Bihar, Uttar Pradesh, West Bengal**, etc.
   - Lead to loss of lives, crop damage, and displacement of people.
3. **Cyclones**
   - India's long coastline (over 7,500 km) is prone to cyclones, especially in the **Bay of Bengal and Arabian Sea**.
   - Examples: Cyclone Fani (2019), Amphan (2020) caused massive destruction.
4. **Droughts**
   - Common in **Rajasthan, Maharashtra, Andhra Pradesh, Karnataka**, due to erratic rainfall.

- Results in crop failure, water scarcity, and farmer distress.

9. **Draw “disaster management cycle.”**

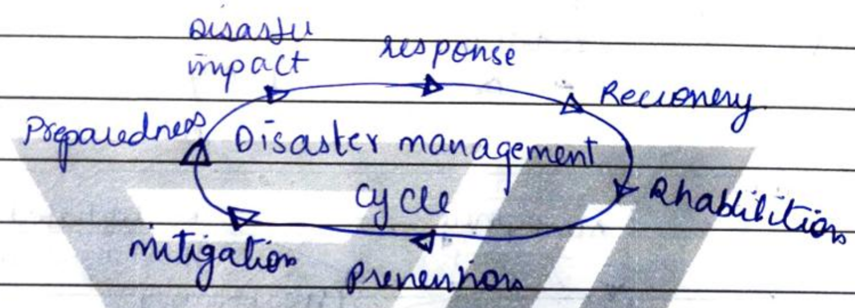


fig: a standard cycle of Disaster Events.

For 5 marks:

1. **Examine the multifaceted components of disaster management including Preparedness, Response, Recovery & Mitigation Measures.**

Q. Examine the multifaceted components of disaster management including Preparedness, Response, Recovery and Mitigation strategies

Disaster management is a systematic process aimed at minimizing the impact of disasters through coordinated efforts before, during, and after an event. It comprises **four key phases: Preparedness, Response, Recovery, and Mitigation.** Each phase plays a distinct role, and together, they form a comprehensive approach to disaster resilience.

1. Preparedness

Definition:
Preparedness involves planning and organizing resources and actions to effectively deal with a disaster before it occurs. (before disaster)

Key Components:

- **Early warning systems**
- **Disaster drills and simulations**
- **Emergency communication plans**
- **Training for first responders and volunteers**
- **Public education and awareness**
- **Stockpiling emergency supplies**

Goal: The main-Aim of preparedness is To ---
To enhance the ability of individuals, communities, and organizations to respond effectively when a disaster strikes.

2. Response

Definition:
Response refers to the immediate actions taken during and right after a disaster to ensure safety, reduce economic losses, and provide emergency assistance. [During disaster] + {Immediately soon after the disaster}

Key Components:

- **Search and rescue operations**
- **Medical aid and trauma care**
- **Evacuation and sheltering**
- **Damage assessment**
- **Coordination among local, national, and international agencies**

Goal: The main goal (Aim) of Response is to---
To save lives, reduce health impacts, and meet the basic needs of the affected population.

3. Recovery

Definition:
Recovery involves restoring affected communities to normal or improved conditions following a disaster. (after disaster)

Key Components:

- **Rebuilding infrastructure (roads, bridges, utilities)**
- **Restoring livelihoods**
- **Psychosocial support and counseling**
- **Economic recovery programs**
- **Long-term health and sanitation improvements**

Goal: The main Aim (Goal) of Recovery is to---
To bring back stability, rebuild better, and reduce future vulnerabilities.

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4. Mitigation

Definition:
Mitigation refers to efforts made to reduce the impact and risks associated with future disasters.

Key Components:

- **Risk assessments and hazard mapping**
- **Enforcing building codes and land-use planning**
- **Climate change adaptation strategies**
- **Ecosystem restoration (e.g., mangroves, wetlands)**
- **Public policies and insurance schemes**

Goal: The main Goal (Aim) of Mitigation is to---
To prevent hazards from becoming disasters or to lessen their potential impacts.

Interrelation of the Four Phases

Four – Phases	Timing	Focus
1. Preparedness:	Before Disaster:	Planning & capacity building
2. Response :→	During Disaster:	Immediate action & relief
3. Recovery:→	After Disaster :	Rehabilitation & reconstruction
4. Mitigation:→	Before & After: (Disaster)	Risk reduction & prevention

All four phases are interconnected and cyclical — effective **mitigation** and **preparedness** reduce the need for **response** and **recovery**, while the lessons from recovery feed into improved preparedness and mitigation.

2. Discuss the role of Growing population, Industrialization, Urbanization & Changing lifestyles in environmental degradation. What measures can be taken to mitigate their impacts?

5M Q. Discuss the Role of Growing-Population, Industrialization, Urbanization & Changing lifestyles in contributing to Environmental degradation. What measures can be taken to mitigate their impact?

May-June: 2025 (10)

What is by GUI&C

Environmental Degradation: Causes and Mitigation

Environmental degradation refers to the **deterioration of the natural environment** due to human activities, leading to issues like pollution, climate change, deforestation, and loss of biodiversity. Several key factors contribute significantly to this problem:

It Causes : ~~...~~ due to following factors:

1. Growing Population

- **Impact:**
  - Increases **demand for land, water, food, and energy.**
  - Leads to **deforestation**, overuse of natural resources, and waste generation.
  - More people = more pressure on limited ecosystems.
- **Example:** Rapid population growth in cities like Delhi or Lagos has led to severe **air and water pollution**, and unmanageable waste.

2. Industrialization

- **Impact:**
  - Industries release **toxic chemicals**, greenhouse gases, and effluents.
  - Promotes **unsustainable resource extraction** (e.g., mining, fossil fuels).
  - Often disregards ecological balance for economic growth.
- **Example:** Industrial zones like **Bhiwandi (India)** or **Wuhan (China)** have faced repeated air and water pollution issues.

3. Urbanization

- **Impact:**
  - Expansion of cities leads to **loss of green cover** and wetlands.
  - Increases **traffic congestion, waste, and air pollution.**
  - Creates **heat islands** and puts pressure on sewage, water, and power systems.
- **Example:** In cities like **Mumbai,** unregulated urban growth has led to frequent flooding and habitat loss.

4. Changing Lifestyle

- **Impact:**
  - Shift towards **consumerism** and **high energy consumption.**
  - Increase in **single-use plastics,** e-waste, fast fashion, and processed food.
  - Overreliance on vehicles, gadgets, and packaged goods worsens pollution.
- **Example:** Fast fashion industries contribute to massive **textile waste and water pollution,** especially in developing countries.

✅ Measures to Mitigate the Impact

Policy and Planning:

- Implement **strict environmental laws** and pollution control norms.(policies)
- Promote **sustainable urban planning** (green buildings, public transport).

Public Awareness and Education:

- Promote **eco-friendly lifestyles** (reduce, reuse, recycle).
- Incorporate **environmental education** in school curricula.

Technology and Innovation:

- Use **clean energy sources** (solar, wind, hydro).
- Adopt **waste management technologies** like composting and recycling.

Population Control:

- Promote **family planning** and **women's education.**
- Strengthen health and reproductive services.

Corporate Responsibility:

- Enforce **green auditing** and sustainability reporting in industries.
- Encourage **CSR initiatives** focusing on the environment.

Conclusion:

To ensure a healthy planet for future generations, it is crucial to **balance development with environmental conservation.** Collective efforts from governments, industries, and individuals are essential to combat (stop) the growing threats of environmental degradation.

combat = try to stop, to fight-against, try to defeat

3. Describe "Paradigm shift" in disaster management. Give an example of paradigm shift in India.

The term *paradigm* means a **model, pattern, or set framework**. In disaster management, a **paradigm shift** refers to the major change in approach, policy, and strategy from the earlier system of handling disasters to a more comprehensive, proactive, and sustainable system.

Earlier Approach (Traditional / Old Paradigm)

- Disaster management was **reactive** in nature.
- Focused mainly on **relief and rescue** after the disaster occurred.
- Government machinery used to provide temporary relief to affected communities.

New Approach (Modern / New Paradigm)

- Disaster management is now **proactive and holistic**.
- Emphasis on **preparedness, prevention, mitigation, and risk reduction** rather than only post-disaster relief.
- Integrates **disaster risk reduction into national development planning**.
- Involves **community participation, international cooperation, and multi-sectoral planning**.
- Supported by policies like the Yokohama Strategy (1994), Hyogo Framework (2005), and Sendai Framework (2015).

Example of Paradigm Shift in India

- Earlier, India's disaster management was handled under the **Ministry of Agriculture** and focused on post-disaster relief.
- After recurrent disasters (e.g., the **2004 Indian Ocean Tsunami**), India realized the need for a new framework.
- In 2005, the **Disaster Management Act** was enacted, leading to the establishment of the **National Disaster Management Authority (NDMA)** under the Ministry of Home Affairs.
- This shifted the focus from relief-centric response to **risk assessment, preparedness, and mitigation**.
- Example initiatives:
  - National Response Plan.
  - Integration of disaster risk reduction into **development planning**.
  - Greater coordination with UN agencies and global strategies.

4. **Discuss various types of Technological disasters & highlight the specific efforts to mitigate such disasters in India.**

Technological disasters are **man-made events** resulting from failure of technology, human error, negligence, or industrial processes. They cause loss of life, property, and environmental damage.

Types of Technological Disasters

1. **Industrial Accidents**
   - Chemical leaks, explosions, fires in factories.
   - Example: **Bhopal Gas Tragedy (1984)**.
2. **Nuclear / Radiological Accidents**
   - Leakage of radioactive materials from nuclear power plants.
   - Example: Concerns after Fukushima disaster (Japan, 2011); India has nuclear plants in Tarapur, Kalpakkam, Kudankulam.
3. **Chemical Disasters**
   - Release of toxic chemicals during storage, transport, or processing.
   - Example: Vizag LG Polymers Gas Leak (2020).

4. **Biological Disasters**
    - Accidental release of viruses/bacteria, lab accidents, epidemics due to human negligence.
5. **Accidents in Transport & Mines**
    - Air crashes, ship accidents (oil spills), train collisions, and mine cave-ins.

Mitigation Efforts in India

- **Legal Framework**:
    - *Factories Act (1948)* – ensures worker safety.
    - *Environment Protection Act (1986)* – after Bhopal tragedy.
    - *Disaster Management Act (2005)* – holistic management of disasters.
- **Institutions**:
    - *NDMA* – issues guidelines for chemical/industrial safety.
    - *AERB* – monitors nuclear power plants.
    - *Crisis Management Group* – coordinates during major industrial/nuclear emergencies.
- **Safety & Preparedness**:
    - Hazardous Chemicals Rules (1989), Chemical Accidents Rules (1996).
    - On-site/off-site emergency plans for industries.
    - Safety audits, inspections, and mock drills.
- **Response Capacity**:
    - *NDRF* has specialized training for CBRN (Chemical, Biological, Radiological, Nuclear) disasters.
    - Community awareness and training programs.

5. **Explain Natural & Manmade disasters. Which disasters can be anticipated & how?**

1. Natural Disasters

- Caused by natural forces beyond human control.
- Examples: Earthquakes, Cyclones, Floods, Droughts, Volcanic Eruptions, Tsunamis.
- Impact: Large-scale loss of life, property, and environment.

2. Man-made (Technological) Disasters

- Result from human negligence, failure of technology, or deliberate actions.
- Examples: Industrial accidents (Bhopal Gas Tragedy 1984), Nuclear accidents, Chemical leaks, Transport accidents, Terrorist attacks.
- Impact: Environmental damage, health hazards, long-term socio-economic effects.

Disasters That Can Be Anticipated

✅ **Natural Disasters** (partially predictable with technology):

- **Cyclones** – predicted using satellites & Doppler radars.
- **Floods** – river gauge monitoring & rainfall forecasting.
- **Droughts** – rainfall patterns & soil moisture analysis.
- **Tsunamis** – ocean sensors & seismic data.
- **Volcanic eruptions** – gas emissions, seismic tremors.

❌ **Difficult-to-Predict Natural Disasters**:

- **Earthquakes** – sudden, no reliable forecasting yet.
- **Landslides** – partly anticipated in high-risk zones, but exact time uncertain.

✅ **Man-made Disasters** (mostly preventable/anticipatable):

- **Industrial & Chemical accidents** – through strict safety audits, monitoring hazardous industries.
- **Transport accidents** – better infrastructure, safety protocols, real-time tracking.
- **Nuclear accidents** – safety drills, radiation monitoring.
- **Terrorist attacks** – intelligence & surveillance systems.

6. **Describe disaster management cycle & also PDCA cycle of disaster management.**

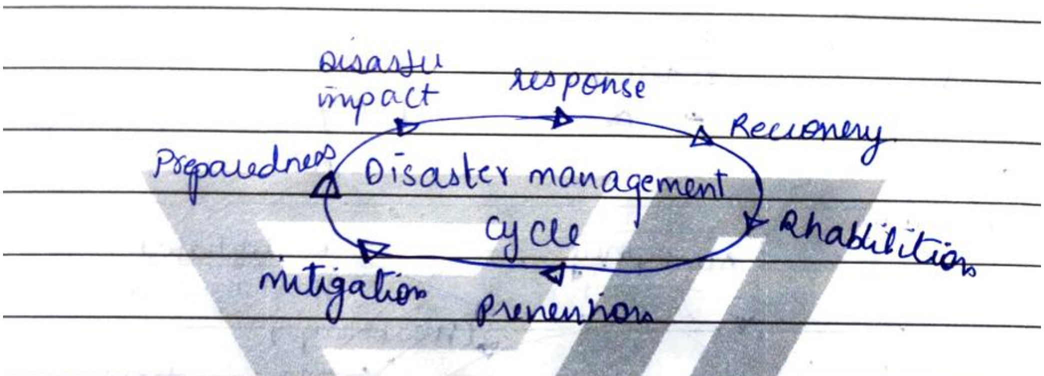


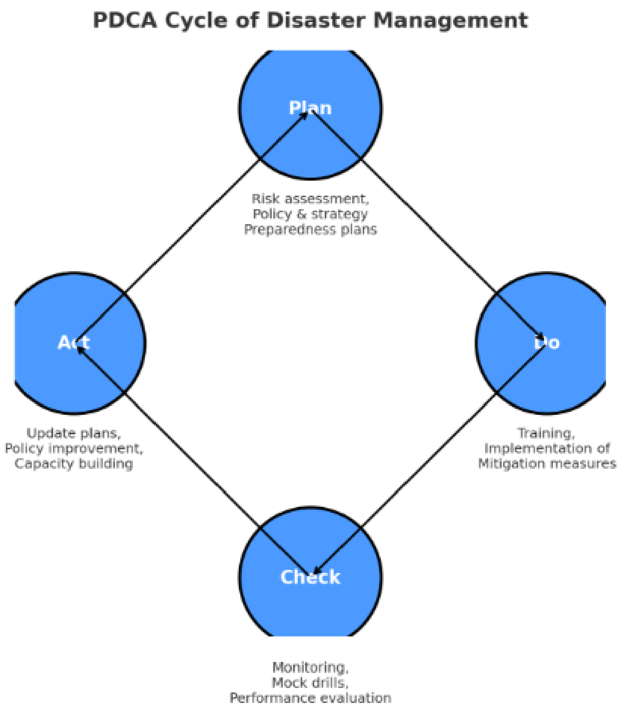
fig 1 a standard cycle of Disaster Events.

1. Disaster Management Cycle

The Disaster Management Cycle describes the continuous process of dealing with disasters before, during, and after they occur.

1. **Preparedness** – It includes planning, training, and awareness programs conducted in advance so that communities and authorities are ready to respond when a disaster strikes.
2. **Impact** – This stage is when the disaster actually occurs, causing immediate damage and disruption to life, property, and environment.
3. **Response** – It refers to emergency actions such as rescue, relief, medical aid, and providing basic needs immediately after the disaster.
4. **Recovery** – Focuses on restoring essential services like power, water, communication, and bringing life back to normal as quickly as possible.
5. **Rehabilitation** – It is the long-term process of rebuilding infrastructure, housing, and livelihoods to help affected people recover completely.
6. **Prevention** – Refers to proactive measures taken to avoid disasters altogether, such as enforcing building codes or banning settlement in flood-prone areas.
7. **Mitigation** – These are efforts aimed at reducing the severity of disasters, for example, constructing embankments to reduce flood impact or earthquake-resistant buildings.

➡️ This forms a **continuous loop** where each phase strengthens the next.



2. PDCA Cycle of Disaster Management

The PDCA (Plan–Do–Check–Act) cycle is a management framework that helps in continuous improvement of disaster management systems.

1. **Plan** – At this stage, hazards and vulnerabilities are identified, risks are assessed, and a detailed disaster management plan is prepared with clear roles and resources.
2. **Do** – The plan is implemented by conducting awareness drives, mock drills, capacity-building programs, and initiating response measures during actual disasters.
3. **Check** – Monitoring and evaluation are done to assess how well the plan worked; for example, analyzing gaps noticed during drills or reviewing the response after a disaster.
4. **Act** – Based on the evaluation, corrective measures are taken to improve future disaster management strategies, ensuring better preparedness and resilience.

➡️ This cycle keeps repeating, ensuring **continuous learning and improvement** in disaster management.